Integrale gamma și beta

• Calculați următoarele integrale, folosind integralele gamma și beta:

1. $I = \int_0^\infty x^5 e^{-2x}\,dx$

$2x = t \Rightarrow x = \frac{t}{2} \Rightarrow dx = \frac{1}{2}dt$

$x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$I = \frac{1}{2^6}\int_0^\infty t^5 e^{-t}\,dt = \frac{1}{2^6}\int_0^\infty e^{-t}t^{6-1}\,dt = \frac{1}{2^6}\Gamma(6) = \frac{5!}{2^6} = \frac{15}{8}$

2. $I = \int_{-\infty}^\infty x^6 e^{-x^2}\,dx$

$f(x) = x^6 e^{-x^2} = (-x)^6 e^{-(-x)^2} = f(-x) \Rightarrow f$ e pară $\Rightarrow I = 2\int_0^\infty x^6 e^{-x^2}\,dx$

$x^2 = t \Rightarrow x = t^{\frac{1}{2}} \Rightarrow dx = \frac{1}{2}t^{-\frac{1}{2}}dt$

$x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$I = \int_0^\infty t^3 e^{-t} t^{-\frac{1}{2}}\,dt = \int_0^\infty t^{\frac{5}{2}} e^{-t}\,dt = \Gamma\left(\frac{7}{2}\right) = \Gamma\left(3+\frac{1}{2}\right) = 1\cdot 3\cdot 5\cdot\frac{\sqrt{\pi}}{2^3} = \frac{15\sqrt{\pi}}{8}$

3. $I = \int_{-1}^\infty \sqrt{x+1}\, e^{-x-1}\,dx$

$x+1 = t \Rightarrow dx = dt$

$x = -1 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$I = \int_0^\infty t^{\frac{1}{2}} e^{-t}\,dt = \Gamma\left(\frac{3}{2}\right) = \Gamma\left(1+\frac{1}{2}\right) = \frac{1}{2}\cdot\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$

4. $I = \int_0^1 \sqrt{x}\,\ln^3 x\,dx$

$\ln x = -t \Rightarrow x = e^{-t} \Rightarrow dx = -e^{-t}dt$

$x = 0 \Rightarrow t = \infty$

$x = 1 \Rightarrow t = 0$

$\Gamma(p) = \int_0^\infty e^{-x}x^{p-1}\,dx$

$I = \int_\infty^0 e^{-\frac{1}{2}t}(-t^3)\cdot(-e^{-t})\,dt = -\int_0^\infty t^3 e^{-\frac{3}{2}t}\,dt$

$\frac{3}{2}t = y \Rightarrow t = \frac{2}{3}y \Rightarrow dt = \frac{2}{3}dy$

$t = 0 \Rightarrow y = 0$

$t = \infty \Rightarrow y = \infty$

$I = -\frac{2^4}{3^4}\int_0^\infty y^3 e^{-y}\,dy = -\frac{2^4}{3^4}\Gamma(4) = -\frac{2^4}{3^4}3! = -\frac{2^5}{3^3} = -\frac{32}{27}$

5. $I = \int_1^\infty \frac{\ln x}{x^a}\,dx$, $a > 1$

$\ln x = t \Rightarrow x = e^t \Rightarrow dx = e^t dt$

$x = 1 \Rightarrow t = 0$; $x = \infty \Rightarrow t = \infty$

$$I = \int_0^{\infty} t\, e^{-t(a-1)}\, dt$$

$t(a-1) = y \Rightarrow t = \frac{1}{a-1} y \Rightarrow dt = \frac{1}{a-1} dy$

$t = 0 \Rightarrow y = 0$

$t = \infty \Rightarrow y = \infty$

$$I = \frac{1}{(a-1)^2} \int_0^{\infty} y\, e^{-y}\, dy = \frac{1}{(a-1)^2} \Gamma(2) = \frac{1}{(a-1)^2}$$

6. $$I = \int_0^1 \sqrt{x - x^2}\, dx = \int_0^1 x^{\frac{1}{2}} (1-x)^{\frac{1}{2}}\, dx = B\left(\frac{3}{2}, \frac{3}{2}\right) = \frac{\Gamma\left(\frac{3}{2}\right) \cdot \Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{3}{2} + \frac{3}{2}\right)} = \frac{\frac{1}{2}\sqrt{\pi} \cdot \frac{1}{2}\sqrt{\pi}}{1!} = \frac{\pi}{8}$$

7. $$I = \int_0^1 \frac{1}{\sqrt[3]{x^2(1-x)}}\, dx = \int_0^1 x^{-\frac{2}{3}} (1-x)^{-\frac{1}{3}}\, dx = B\left(\frac{1}{3}, \frac{2}{3}\right) = \frac{\Gamma\left(\frac{1}{3}\right) \cdot \Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{1}{3} + \frac{2}{3}\right)} = \Gamma\left(\frac{1}{3}\right) \cdot \Gamma\left(1 - \frac{1}{3}\right) =$$

$$= \frac{\pi}{\sin\frac{\pi}{3}} = \frac{2\pi}{\sqrt{3}} = \frac{2\sqrt{3}\pi}{3}$$

8. $$I = \int_{-\infty}^{\infty} e^{2x} (1 + e^{2x})^{-2}\, dx$$

$e^{2x} = t \Rightarrow e^{2x}\, dx = \frac{1}{2} dt$

$x = -\infty \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$$I = \frac{1}{2} \int_0^{\infty} (1+t)^{-2}\, dt = \frac{1}{2} \int_0^{\infty} \frac{t^0}{(1+t)^2}\, dt = \frac{1}{2} B(1,1) = \frac{1}{2} \frac{\Gamma(1) \cdot \Gamma(1)}{\Gamma(2)} = \frac{1}{2}$$

9. $$I = \int_1^e \frac{1}{x} \ln^3 x\, (1 - \ln x)^4\, dx$$

$\ln x = t \Rightarrow \frac{1}{x} dx = dt$

$x = 1 \Rightarrow t = 0$

$x = e \Rightarrow t = 1$

$$I = \int_0^1 t^3 (1-t)^4\, dt = B(4,5) = \frac{\Gamma(4) \cdot \Gamma(5)}{\Gamma(9)} = \frac{3! \cdot 4!}{8!} = \frac{1}{5 \cdot 7 \cdot 8} = \frac{1}{280}$$

10. $$I = \int_0^{\infty} \frac{x}{1 + x^6}\, dx$$

$x^6 = t \Rightarrow x = t^{\frac{1}{6}} \Rightarrow dx = \frac{1}{6} t^{-\frac{5}{6}} dt$

$x = 0 \Rightarrow t = 0$

$x = \infty \Rightarrow t = \infty$

$$I = \int_0^{\infty} \frac{t^{\frac{1}{6}}}{1+t} \cdot \frac{1}{6} t^{-\frac{5}{6}}\, dt = \frac{1}{6} \int_0^{\infty} \frac{t^{-\frac{2}{3}}}{1+t}\, dt = \frac{1}{6} B\left(\frac{1}{3}, \frac{4}{3}\right) = \frac{\sqrt{3}\pi}{9}$$

11. $$I = \int_{-4}^{3} \frac{1}{\sqrt[6]{(x+4)(3-x)^5}}\,dx$$

$\frac{x+4}{7} = t \Rightarrow dx = 7\,dt$

$x = -4 \Rightarrow t = 0$

$x = 3 \Rightarrow t = 1$

$$I = \int_0^1 7\cdot (7t)^{-\frac{1}{6}}\cdot(7(1-t))^{-\frac{5}{6}}\,dt = \int_0^1 t^{-\frac{1}{6}}(1-t)^{-\frac{5}{6}}\,dt = B\left(\frac{5}{6},\frac{1}{6}\right) = \frac{\Gamma\left(\frac{5}{6}\right)\cdot\Gamma\left(\frac{1}{6}\right)}{\Gamma(1)} = \frac{\pi}{\sin\frac{\pi}{6}} = 2\pi$$

12. $$I = \int_0^1 \frac{1}{\sqrt[n]{1-x^m}}\,dx$$

$x^m = t \Rightarrow x = t^{\frac{1}{m}} \Rightarrow dx = \frac{1}{m}t^{\frac{1}{m}-1}\,dt$

$x = 0 \Rightarrow t = 0$

$x = 1 \Rightarrow t = 1$

$$I = \frac{1}{m}\int_0^1 (1-t)^{-\frac{1}{n}} t^{\frac{1}{m}-1}\,dt = \frac{1}{m}B\left(\frac{1}{m},1-\frac{1}{n}\right) = \frac{1}{m}\frac{\pi}{\sin\frac{\pi}{m}}$$